



Year 11 Mathematics

June 2011

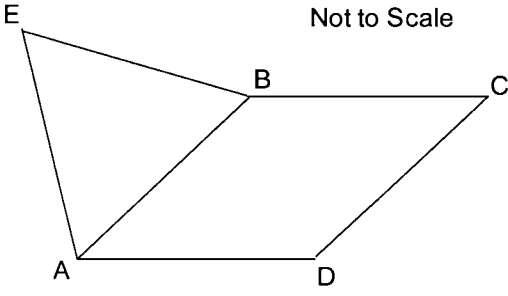
Time: 70 Minutes

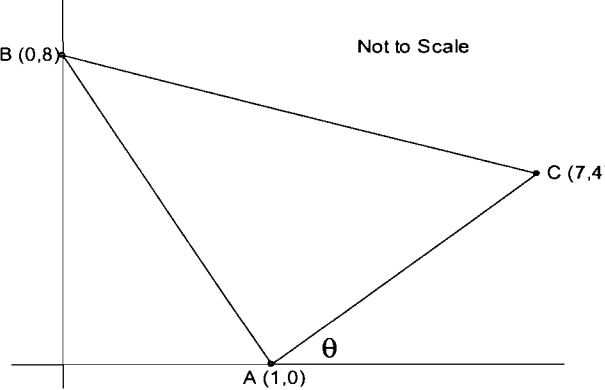
DIRECTIONS

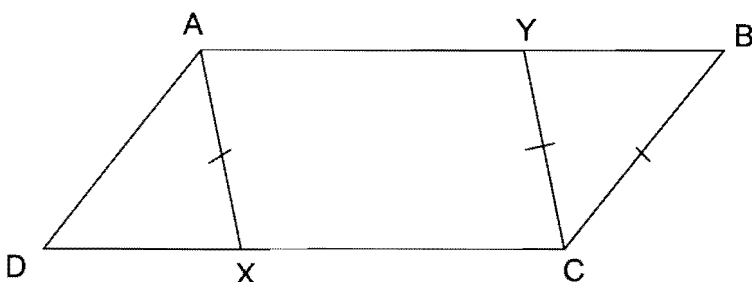
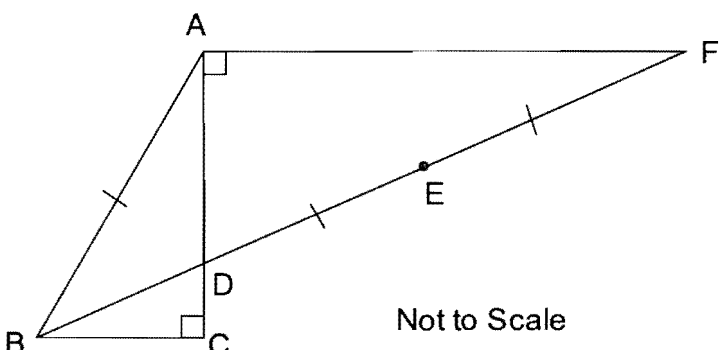
- Full working should be shown in every question. Marks may be deducted for careless or badly arranged work.
- Use black or blue pen only (*not pencils*) to write your solutions.
- No liquid paper is to be used. If a correction is to be made, one line is to be ruled through the incorrect answer.

1. a) Find the domain and range of $y = 3\sqrt{2x - 5}$. 2
- b) If $f(x) = x^2 + x$, determine if $f(x)$ is odd, even, or neither. Show all working. 2

2. a) Sketch the region which satisfies $x^2 + y^2 < 1$ 3
- b) State the domain and range of the region. 3

3.  ABCD is a rhombus.
ABE is an equilateral triangle.
Given $\angle BCD = 48^\circ$.
- Find:
- i) $\angle EAD$, giving reasons for your answer. 2
- ii) $\angle EDA$, giving reasons for your answer. 2

4.  The points A, B, C have co-ordinates (1,0), (0,8), and (7,4) as shown.
The angle between line AC and the x-axis is θ .
- i) Find the size of angle θ in degrees. 1
- ii) Find the equation of line AC. 1
- iii) Find co-ordinates of D, the midpoint of AC. 1
- iv) Show that AC is perpendicular to BD. 2
- v) What does part (iv) show about $\triangle ABC$? 1
- vi) Find the area of $\triangle ABC$. 2
- vii) Write down the co-ordinates of E such that ABCE is a rhombus. 1

5.	 $ABCD$ is a parallelogram $AX = CY = BC$ i) Prove $\triangle AXD \equiv \triangle BYC$ ii) Hence, prove $AYCX$ is a parallelogram.	3
6.	A function is defined as $f(x) = \begin{cases} x & \text{for } x < -1 \\ x^2 - 2 & \text{for } -1 \leq x < 1 \\ x^2 & \text{for } x \geq 1 \end{cases}$ Evaluate a) $f(-3) + f(0) + f(1)$ b) $f(a^2 + 1)$	1 1
7.	a) Sketch the region $y \leq \sqrt{9 - x^2}$ b) Find the domain and range of $y = \sqrt{x^2 - 25}$	2 2
8.	a) Show that the line $3x + 4y + 25 = 0$ is a tangent to the circle $x^2 + y^2 = 25$ b) Find the equation of the line through $(a \cos \theta, a \sin \theta)$ perpendicular to $x \cos \theta + y \sin \theta = a$	2 3
9.	 Not to Scale In the diagram $AC \perp BC$, $AC \perp AF$ and $AB = DE = EF$ i) Show $\angle DBC = \angle DFA$ ii) G is on the line AF such that $EG \parallel AC$ Show $\triangle AGE \equiv \triangle FGE$ iii) Prove $\angle ABD = 2\angle DBC$	2 3 2
~ END OF EXAM ~		

Question 1.

a) $y = 3\sqrt{2x-5}$

ID: $x \geq \frac{5}{2}$ R: $y \geq 0$

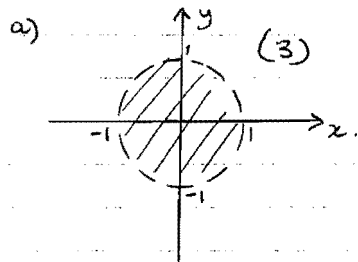
b) $f(x) = x^2 + x$ $f(-x) = (-x)^2 + (-x)$
 $= x^2 - x$

test even: $f(x) = f(-x)$
 $x^2 + x \neq x^2 - x$ NOT EVEN

test odd: $f(x) = -f(-x)$
 $x^2 + x = -(x^2 - x)$
 $\neq -x^2 + x$ NOT ODD

$\therefore f(x)$ is neither

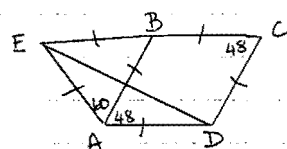
Question 2



b) ID: $-1 < x < 1$
 R: $-1 < y < 1$

Question 3.

(a possible response)



i) $\angle BAD = \angle BCD$ (opposite $\angle$ s equal in rhombus) $\angle BAD = 48^\circ$
 $\angle EAB = 60^\circ$ ($\triangle ABE$ equilateral given)
 $\therefore \angle EAD = 48 + 60 = 108^\circ$

ii) since $EA = AB$ (sides of equilateral $\triangle$)
 and $AB = AD$ (sides of rhombus)
 then $EA = AD$

$\angle EDA = \angle AED$ (equal $\angle$ s opposite equal sides)
 $= \frac{180 - 108}{2} = 36^\circ$

Question 4

i) $m = \frac{y_2 - y_1}{x_2 - x_1}$

$= \frac{4-0}{7-1}$

$= \frac{2}{3}$

$\tan \theta = m$

$= \frac{2}{3}$

$\theta \approx 34^\circ$

ii) $y - y_1 = m(x - x_1)$

$y - 0 = \frac{2}{3}(x - 1)$

$3y = 2x - 2$
 $2x - 3y - 2 = 0$

iii) D: $(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2})$

$= (\frac{1+7}{2}, \frac{0+4}{2})$

$= (4, 2)$

iv) $m_1 = \frac{2}{3}$

$m_2 = \frac{y_2 - y_1}{x_2 - x_1}$

$= \frac{8-2}{0-4}$

$= -\frac{3}{2}$

$\therefore m_1 \times m_2 = \frac{2}{3} \times -\frac{3}{2} = -1$

$AC \perp BD$

(could use perp dis formula.)

v) isosceles $\triangle$

vi) length $AC = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$
 $= \sqrt{(7-1)^2 + (4-0)^2}$

$= \sqrt{52}$ units

D BD $= \sqrt{(0-4)^2 + (8-2)^2}$
 $= \sqrt{52}$

Area $\triangle ABC = \frac{1}{2} \times \sqrt{52} \times \sqrt{52}$
 $= 26$ u²

vii) Mid pt $AC = (4, 2)$

Mid pt BE
 $4 = \frac{x+0}{2}, 2 = \frac{8+y}{2}$
 $(8, -4)$

9

Question 5.

i) (a possible response)

(3) marks

In $\triangle AXB$ and $\triangle BYC$

$\hat{A}DX = \hat{Y}BC$ (opposite $\angle$ s in parallelogram equal)

$AD = BC$ (opposite sides in parallelogram equal)

$AX = CY$ (given)

$\therefore AD = AX = BC = CY$

$\triangle ADX$ is isosceles (2 sides equal)

then $\angle AXD = \angle ADX$ (base $\angle$ s, isosceles $\triangle$)

Similarly $\triangle BYC$ is isosceles

and $\angle CYB = \angle CBY$ (base $\angle$ s, isosceles $\triangle$)

$= \angle ADX$ (from above)

$\therefore \triangle AXB \equiv \triangle BYC$ (AAS)

ii) Since $AB = DC$ (opposite sides in parallelogram)

and $DX = BY$ (matching sides in congruent $\triangle$ s)

then $AY = AB - YB$
and $CX = CD - DX$ } (subtraction of sides)

$\therefore AY = CX$

and $AX = CY$ (given)

$\therefore AYCX$ is a parallelogram (2 pairs of equal sides)

Question 6.

a) $f(-3) = -3$ $f(0) = -2$ $f(1) = 1$

$\therefore f(-3) + f(0) + f(1) = -3 - 2 + 1$

$= -4$

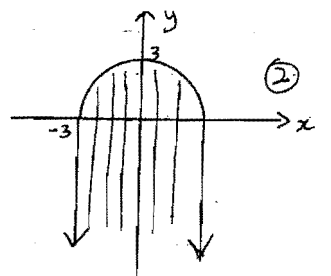
(1)

b) $f(a^2+1) = (a^2+1)^2$
 $= a^4 + 2a^2 + 1$

(1)

Question 7.

a)



(2)

b) $y = \sqrt{x^2 - 25}$
 $x^2 - 25 \geq 0$

D: $x \leq -5, x \geq 5$ (1)

R: $y \geq 0$ (1)

Question 8.

a) tangent to circle - perp dist = radius = 5
centre = (0,0)

$$\begin{aligned} \text{perp d} &= \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}} \\ &= \frac{|3 \times 0 + 4 \times 0 + 25|}{\sqrt{9 + 16}} \\ &= \frac{25}{5} = 5 \end{aligned} \quad (1)$$

$\therefore$ since line is perpendicular to radius it is a tangent (1)

b) $x \cos \theta + y \sin \theta = a$

$$y \sin \theta = a - x \cos \theta$$

$$y = \frac{a}{\sin \theta} - \frac{\cos \theta}{\sin \theta} x$$

$$m_1 = -\frac{\cos \theta}{\sin \theta} \quad \text{now } m_2 = -\frac{1}{m_1} = \frac{\sin \theta}{\cos \theta} \quad (1)$$

$$y - y_1 = m(x - x_1)$$

$$y - a \sin \theta = \frac{\sin \theta}{\cos \theta} (x - a \cos \theta)$$

$$\cos \theta y - a \sin \theta \cos \theta = \sin \theta x - a \sin \theta \cos \theta$$

$$\left. \begin{aligned} y &= \frac{\sin \theta}{\cos \theta} x \\ y &= \tan \theta x \end{aligned} \right\} (1)$$

Question 9 (possible solutions)

i) aim: show $\angle DBC = \angle DFA$.

Method: $\angle ADF = \angle CDB$ (vertically opposite $\angle$ s)

$$\angle CAF = \angle ACB \text{ (given - b)} \quad (2)$$
$$= 90^\circ$$

$$\therefore \angle AFD = \angle DBC \text{ (L sum of } \triangle \text{s)}$$

ii) aim: to prove $\triangle AGE \cong \triangle FGE$

method: $\angle CAG = \angle EGF$ (corresponding $\angle$ s, $AC \parallel GE$)

$$= 90^\circ$$

$$\angle AGE = 180 - \angle EGF \text{ (L sum straight line)}$$
$$= 90^\circ$$

Since $EF = ED$ (given) (3)

then $EF : ED = 1 : 1$

$FQ : QA = 1 : 1$ (Intercepts on $\parallel$ lines)

$$FQ = QA$$

EQ is common

$$\therefore \triangle AGE \cong \triangle FGE \text{ (SAS)}$$

iii) aim: prove $\angle ABD = 2 \times \angle DBC$

method: let $\angle DBC = x$

$$\therefore \angle BFG = x = \angle GAE \text{ (matching } \angle \text{s in congruent } \triangle \text{s)}$$

$$\angle AED = 2x \text{ (External } \angle \triangle \text{ equals sum of interior opposite } \angle \text{s)}$$

$$EF = AE \text{ (matching sides } \cong \triangle \text{s)} \quad (2)$$

$$\therefore AB = AE$$

$$\angle ABE = 2x \text{ (base } \angle \text{s in isosceles } \triangle \text{ equal)}$$
$$= 2 \times \angle DBC.$$